

Warm-Up Quiz — Day 11

Taylor Expansions and the Delta Method

Complete before lecture. 10 minutes.

Name: _____

Today we apply the CLT to prove OLS is asymptotically normal, and use the delta method to find the distribution of functions of estimators. The key tool is the first-order Taylor expansion: $f(x) \approx f(a) + f'(a)(x - a)$.

1. First-Order Taylor Expansion.

Write the linear approximation $f(x) \approx f(a) + f'(a)(x - a)$ for each function, expanded around the given point.

(a) $f(x) = e^x$ around $a = 0$.

(b) $f(x) = \log(x)$ around $a = 1$.

(c) $f(x) = 1/x$ around $a = 2$.

2. Delta Method (Scalar Case).

If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then:

$$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, [g'(\theta)]^2 \sigma^2)$$

(a) Suppose $\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N(0, \sigma^2)$. Find the asymptotic distribution of $\hat{\mu}^2$.
(Hint: $g(\mu) = \mu^2$, so $g'(\mu) = ?$)

(b) Suppose $\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1 - p))$. Find the asymptotic distribution of $\log(\hat{p}/(1 - \hat{p}))$.
(Hint: $g(p) = \log(p/(1 - p))$, so $g'(p) = 1/(p(1 - p))$.)

3. Combining Slutsky and CLT (Preview).

We showed $\hat{\beta} - \beta = \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1} \left(\frac{1}{n}\mathbf{X}'\mathbf{e}\right)$. Suppose:

- $\frac{1}{n} \mathbf{X}' \mathbf{X} \xrightarrow{p} \mathbf{Q}$ (a constant matrix), and
- $\frac{1}{\sqrt{n}} \mathbf{X}' \mathbf{e} \xrightarrow{d} N(\mathbf{0}, \mathbf{\Omega})$.

(a) Rewrite $\sqrt{n}(\hat{\beta} - \beta)$ in terms of $(\frac{1}{n} \mathbf{X}' \mathbf{X})^{-1}$ and $\frac{1}{\sqrt{n}} \mathbf{X}' \mathbf{e}$.

(b) By Slutsky's theorem, what is the asymptotic distribution of $\sqrt{n}(\hat{\beta} - \beta)$?

4. Jacobian for the Multivariate Delta Method.

If $\boldsymbol{\beta} = (\beta_1, \beta_2)'$ and $g(\boldsymbol{\beta}) = \beta_1/\beta_2$, compute the Jacobian row vector $\nabla g = \left(\frac{\partial g}{\partial \beta_1}, \frac{\partial g}{\partial \beta_2} \right)$.

Answer Key — Day 11

1. (a) $e^x \approx 1 + x$.
 (b) $\log(x) \approx 0 + 1 \cdot (x - 1) = x - 1$.
 (c) $1/x \approx 1/2 + (-1/4)(x - 2) = 1 - x/4$.

2. (a) $g'(\mu) = 2\mu$, so $\sqrt{n}(\hat{\mu}^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$.
 (b) $g'(p) = 1/(p(1-p))$, so $\sqrt{n}(\hat{g} - g) \xrightarrow{d} N\left(0, \frac{1}{p(1-p)}\right)$. (The $p(1-p)$ in the variance of $\hat{p}$ and the $[p(1-p)]^{-2}$ from the delta method partially cancel.)

3. (a) $\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1} \frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{e}$.
 (b) By Slutsky: $\left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1} \xrightarrow{p} \mathbf{Q}^{-1}$ and $\frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{e} \xrightarrow{d} N(\mathbf{0}, \mathbf{\Omega})$, so

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{Q}^{-1}\mathbf{\Omega}\mathbf{Q}^{-1}).$$

4. $\nabla g = \left(\frac{1}{\beta_2}, -\frac{\beta_1}{\beta_2^2}\right)$.